Bengaluru, Karnataka, 560114
8904295930
aontrol@gmail.com
aontrol
Aontrol

Abhirama M

Professional Summary

Hardware Developer and Electronics Engineer with over 6 years of continuous, hands-on experience in digital electronics, KiCad PCB design, and end-to-end embedded systems development. Proven track record of designing, assembling, and delivering commercial-grade custom electronic products independently. Highly proficient in bare-metal firmware development (ARM Cortex-M, ESP32, RP2040), direct register-level programming, Linux system administration, and rigorous component-level debugging. Possesses additional working knowledge of industrial automation (PLC/HMI/SCADA) and FPGA digital design (System Verilog), enabling comprehensive support across the entire hardware engineering and manufacturing lifecycle.

Technical Skills

Expertise (Hardware) KiCad, Schematic Capture, PCB Layout, SMD/Through-Hole Soldering, Multisim, Oscilloscope, Logic Analyzer, Signal Generator, Digital Multimeter (DMM) Diagnostics, Component-Level Circuit Debugging, Signal Integrity, MOSFET Switching Analysis.

Expertise (Embedded) ARM Cortex-M (M0+/M3), RP2040, ESP32, LPC1768, FreeRTOS, I2C, SPI, UART, EEPROM, LoRa, RF, WiFi, Bluetooth, ADC/DAC.

Expertise (SysAdmin) Embedded C, Assembly, Register-Level Programming, C++, Python, MicroPython, RTOS. Advanced Linux Administration (Ubuntu Server, Mint, Manjaro), Custom Makefile Build Systems, Docker Containerization, VirtualBox, Bash Scripting.

Working Knowledge System Verilog, Verilog, FPGA Development, Intel FPGA Starter Edition (ModelSim), PLC Programming, Human-Machine Interface (HMI), SCADA.

Professional Experience

2020 – Present **Freelance Hardware Developer**, *Independent Consultant*

- Engineered and manufactured custom embedded solutions and consumer-ready electronic devices based on highly specific client requirements.
- Managed the complete end-to-end product lifecycle, including requirement gathering, KiCad PCB layout, component sourcing, manual soldering assembly, and final firmware deployment.
- Developed highly optimized bare-metal firmware utilizing direct register-level programming to bypass standard abstraction layers, ensuring strict timing requirements and custom peripheral control.
- Performed rigorous component-level debugging utilizing oscilloscopes, signal generators, and digital multimeters to diagnose communication faults, ensure stable analog routing, and validate circuit integrity.
- Successfully delivered and commercialized physical devices, solving practical, real-world problems through custom digital electronics and microcontroller integration.

Oct 2025 – Aug 2026 **Embedded Systems Intern**, *Cranes Varsity Pvt Ltd*

- Developed embedded applications using Embedded C on the LPC1768 (ARM Cortex-M) architecture using the Keil IDE.
- Programmed microcontrollers for complex peripheral interfacing, effectively managing UART communication, ADC conversions, and GPIO controls for hardware debugging.

Key Projects

Intelligent POS System, *Embedded C, FreeRTOS, LPC1768, Keil IDE*

- Engineered a bare-metal Point of Sale (POS) system on the ARM Cortex-M based LPC1768 architecture using Embedded C and the Keil IDE.
- Implemented FreeRTOS for efficient task scheduling, managing concurrent processes including GPIO polling, LCD interfacing, and switch debouncing.
- Ensured strict real-time constraints and optimized dynamic memory allocation for continuous, reliable system uptime.

Universal Electronic Meter, *RP2040, MicroPython, Hardware Design*

- Designed a compact, multi-purpose measuring device utilizing a 10 mA constant current source and precise ADC voltage drop measurements for an autoranging ohmmeter circuit.
- Developed custom hardware interfaces and measurement algorithms for signal input, voltage, and current sensing using the Raspberry Pi Pico and a custom MicroPython environment.

PumpPilot (IoT Control System), *ESP32, LoRa, Bluetooth, KiCad*

- Built an ESP32-based automatic pump control system featuring timer-based operation, non-volatile memory setting storage, and long-range LoRa communication.
- Integrated a Bluetooth module for Android app connectivity and designed the physical motor control and relay switching circuits for reliable, application-level deployment.

Linux SysAdmin & Automation, *Bash, Makefiles, Docker*

- Administered local Linux environments (Ubuntu/Mint) and provisioned VirtualBox VMs to create robust, isolated environments for firmware compilation and hardware simulation.
- Developed custom Bash and Python automation tools, integrated with Makefile build systems, to optimize C code modularization and eliminate manual repetitive compilation tasks.

Education

2023 – 2026 **Bachelor of Engineering in Electronics and Communication**, *Kalpataru Institute of Technology*, Tiptur, CGPA: 7.56

2020 – 2023 **Diploma in Electronics and Communication**, *Sanjay Memorial Polytechnic*, Sagara, CGPA: 9.26